

Physics Mock Test

Electromagnetic Induction

Class 12 | UNIT 4

Marks: 24

Time: 60 Minutes

Coverage: Faraday's Law, Lenz's Law, Motional EMF, Eddy Currents, Inductance (Self/Mutual), and Transformers.

Instructions: For Section 2, use $\pi \approx 3.14$ where applicable. Show the units in your final answer.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Faraday's Discovery

According to Faraday's law of induction, the magnitude of the induced electromotive force (EMF) in a circuit is directly proportional to:

- a) The rate of change of magnetic flux
- b) The total magnetic flux linked
- c) The steady current in the coil
- d) The total resistance of the coil

2. Significance of Lenz's Law

Lenz's law is a consequence of the law of conservation of energy. It states that the direction of induced current is such that it:

- a) Opposes the change in magnetic flux
- b) Enhances the external magnetic field
- c) Opposes the flow of charge
- d) Cancels out the resistance

3. Motional EMF Expression

For a conductor of length l moving with a velocity v at an angle θ to a uniform magnetic field B , the induced EMF is:

- a) $B l v$
- b) $B l v \sin\theta$
- c) $B l^2 v$
- d) $B v / l$

4. Eddy Currents

Eddy currents (Foucault currents) are induced in bulk pieces of conductors when they are subjected to:

- a) Constant DC current
- b) Changing magnetic flux
- c) A constant electric field
- d) High temperature only

5. SI Unit of Inductance

The Henry (H) is the SI unit of both self and mutual inductance. One Henry is equivalent to:

- a) Volt-second / Ampere
- b) Tesla-meter²
- c) Weber / Second
- d) Joule / Coulomb

6. Transformer Principles

In a step-up transformer, the secondary voltage is higher than the primary voltage because:

- a) $N_s < N_p$
- b) $N_s = N_p$
- c) $N_s > N_p$
- d) Secondary resistance is zero

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Calculation of Induced EMF

A coil of 100 turns has a cross-sectional area of 0.01 m^2 . If the magnetic field perpendicular to the coil changes uniformly from 0.1 T to 0.2 T in 0.5 seconds , calculate the induced EMF.

8. Faraday & Lenz Application

A 50-turn coil experiences a flux change of 0.005 Wb over a time interval of 0.1 s . Calculate the magnitude of induced EMF. Also, state the direction of current if a North pole approaches the coil face.

9. Motional EMF in a Rod

A metallic rod of length 0.5 m moves at a constant speed of 10 m/s perpendicular to a uniform magnetic field of 0.2 T . Determine the EMF induced between the ends of the rod.

10. Self-Inductance of a Solenoid

A solenoid has 500 turns, length 20 cm, and a cross-sectional area of 10 cm^2 . Calculate its self-inductance.

($\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$)

11. Mutual Induction

Two coils have a mutual inductance of 0.01 H. If the current in the primary coil changes from 5 A to 3 A in 0.1 s, calculate the EMF induced in the secondary coil.

12. Transformer Voltage & Power

A transformer has 200 turns in the primary and 4000 turns in the secondary. Calculate the output voltage if 220 V is applied to the primary. Why is high-voltage transmission preferred for power grids?

Answer Key & Solutions

1. a) Rate of change of magnetic flux

$$\varepsilon = -N (d\Phi/dt).$$

2. a) Opposes the change in magnetic flux

Ensures energy conservation.

3. b) $B l v \sin\theta$

Maximum when motion is perpendicular ($\theta=90^\circ$).

4. b) Changing magnetic flux

Creates circulating currents in the bulk metal.

5. a) Volt-second / Ampere

$$L = \varepsilon / (dI/dt).$$

6. c) $N_s > N_p$

Voltage ratio follows turns ratio ($V_s/V_p = N_s/N_p$).

7. 0.2 V

$$\varepsilon = N \cdot A \cdot (\Delta B/\Delta t) = 100 \times 0.01 \times (0.1/0.5) = 1 \times 0.2 = 0.2 \text{ V}.$$

8. 2.5 V | Anticlockwise

$\varepsilon = N (\Delta\Phi/\Delta t) = 50 \times (0.005/0.1) = 2.5 \text{ V}$. As N approaches, the coil face becomes a North pole (Anticlockwise current) to repel it.

9. 1.0 V

$$\varepsilon = Blv = 0.2 \times 0.5 \times 10 = 1.0 \text{ V}.$$

10. $1.57 \times 10^{-3} \text{ H}$ (1.57 mH)

$$L = \mu_0 N^2 A / l = (4\pi \times 10^{-7} \times 500^2 \times 10^{-3}) / 0.2 = (4\pi \times 10^{-7} \times 250000 \times 0.001) / 0.2 \approx 1.57 \text{ mH}.$$

11. 0.2 V

$$\varepsilon = M (\Delta I / \Delta t) = 0.01 \times (2/0.1) = 0.2 \text{ V}.$$

12. 4400 V

$V_s = V_p(N_s/N_p) = 220 \times (4000/200) = 4400 \text{ V}$. High voltage transmission reduces I^2R power losses by reducing current.

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